

CAUSALITY AND FACTOR INVESTING: A PRIMER

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ABSTRACT

Factor investing is a foundational paradigm in quantitative asset management. Yet, despite the proliferation of factors and widespread institutional adoption, most strategies have failed to live up to their in-sample promise. While p -hacking and backtest overfitting have received considerable blame, a more insidious source of error is rarely discussed: the uncritical application of an econometric canon that ignores causal structure. This paper introduces the concept of the factor mirage—a factor model that appears statistically valid but is causally misspecified. We show how collider bias and confounder bias, when embedded in the standard regression framework, can yield misleading inferences, poor out-of-sample performance, and misguided investment decisions. By shifting from associational to causal reasoning, practitioners can build more robust strategies, reduce false discoveries, and restore trust in factor-based approaches.

Keywords: Causal inference, causal discovery, confounder, collider, factor investing, p -hacking, underperformance, systematic losses.

JEL Classification: G0, G1, G2, G15, G24, E44.

AMS Classification: 91G10, 91G60, 91G70, 62C, 60E.

1. INTRODUCTION

Factor investing was once heralded as the future of systematic asset management. Academic breakthroughs like Fama and French’s three-factor model gave rise to a burgeoning field, with hundreds of anomalies proposed and institutionalized (Harvey et al. [2016]). Multi-factor products now manage trillions of dollars in AUM, promising style tilts that enhance returns, reduce risk, or both. Yet, the real-world long-term performance of most such strategies has fallen short of expectations (López de Prado and Zoonekynd [2024]).

This shortfall has prompted soul-searching across the industry. Critics cite *p*-hacking and backtest overfitting as the main culprits, see Bailey et al. [2014], Fabozzi and López de Prado [2018], Harvey and Liu [2020]. Others argue that market participants arbitrage away these opportunities shortly after publication (McLean and Pontiff [2016]). Still some suggest that factors only work in certain regimes, and the recent regime has been unfavorable –a dubious ex-post argument given that the original publications made no regime distinctions (Evans [1994], Anderson [2011]).

These explanations may contain elements of truth, but they overlook a deeper problem: association does not imply causation. The econometric canon applied in factor investing studies—linear regression, two-pass estimation, *p*-values, and correlation-based statistics—rarely discusses causality. Yet investment decisions are inherently causal, because they require the attribution of returns to risk sources. We do not merely want to know if a portfolio composed of high book-to-market stocks delivers positive returns; in order to optimize a portfolio, we need to determine how much of that performance is attributable to (i.e., caused by) the value factor, to the exclusion of (i.e., controlling for) other explanations, and what may disrupt this relationship (i.e., causal mechanism) going forward.

The distinction between association and causation goes beyond semantics. Some associational models may produce good forecasts, at the expense of offering no risk and return attribution, thus exposing investors to unknown or unwanted risks. Without accounting for causal structure, models are likely to be biased, unstable, and unprofitable out-of-sample.¹

2. FROM THE “FACTOR ZOO” TO THE “FACTOR MIRAGE”

Cochrane [2011] coined the term “factor zoo” to illustrate the explosion of empirical findings in asset pricing. Hundreds of published anomalies now compete for attention, yet most fail to survive replication or implementation. The literature has responded with tools to mitigate data-snooping, such as corrections for multiple testing, the deflated Sharpe ratio, Bayesian shrinkage, and out-of-sample tests (Bailey and López de Prado [2014], López de Prado [2018, 2020]).

¹ Causality has a long history in economics going back to Smith [1776], however this term is often used with different and confusing meanings. For example, Haavelmo [1944] discussed causality in the *ceteris paribus* sense, which simulates a controlled experiment. In contrast, Granger [1969, 1980] discussed causality in a predictive (associational) sense. For Granger, a variable *X* causes *Y* if *X* has predictive power over *Y* after controlling for lags of *Y*. This is not strict causality, because it does not contemplate counterfactuals (Pearl [2009], Pearl et al. [2016]). For example, a rooster Granger-causes the sun to rise, however a controlled experiment will show that the sun will also rise in the absence of a rooster. Modern economists working on (counterfactual) causality include Guido Imbens, Joshua Angrist and David Card, however these concepts have not yet permeated financial economics, where associational statistics is still ubiquitous.

While useful, these techniques address the selection of models, not their specification. A model can be *p*-hacking-free and still misspecified. In this paper, we introduce the concept of the “factor mirage”: an empirical finding that appears sound by conventional statistical standards but is structurally invalid because it misrepresents the causal relationships among variables, leading to a biased risk and return attribution.

Factor mirages arise from two common specification errors: (1) Confounder bias – failing to control for variables that are *causes* of both an independent variable (factor) and the dependent variable (returns); and (2) collider bias – controlling for variables that are *consequences* of both an independent variable and the dependent variable, introducing non-causal associations. In econometric terms, these biases distort coefficient estimates. In financial terms, they lead to inefficient investments, and even systematic losses.

Unlike brute-force *p*-hacking, which is an acknowledged malpractice, the factor mirage is subtler. It arises from practices that are widely taught, widely applied, and rarely questioned.

3. WHERE THE CANON FAILS: ECONOMETRICS WITHOUT CAUSALITY

The econometric methods most widely used in empirical finance—OLS regressions, stepwise model selection, and significance testing—assume that the regression model is correctly specified, and residuals are stationary and behave like white noise. However, these assumptions are often violated in financial applications, particularly in asset pricing models.

A case in point is the assumption of correct model specification. In asset pricing, the standard practice for identifying a factor is to follow a two-pass regression approach: First, run time series regressions of assets excess returns on a set of factors to estimate factor loadings (exposures). Second, run cross-sectional regressions of assets excess returns on the estimated factor exposures to estimate factor premia. The choice of specification is typically driven by associational power maximization, not causal considerations.

A confounder is a variable that is a cause to both an explanatory variable and the dependent variable. Confounder bias arises when the model’s specification does not control for a confounder.² If leverage, for instance, influences both book-to-market and returns, and is not included in the model, the estimated coefficient for book-to-market will be biased in magnitude and perhaps sign. Confounder bias exposes investors to unwanted risks and premia.

A collider is a variable that is causally downstream of both an explanatory variable and the dependent variable. Controlling for a collider introduces a non-causal correlation, which biases coefficient estimates, inflates the adjusted R-squared and tends to lower *p*-values. For example, if quality is influenced by both book-to-market and returns, including it as a control will bias the estimated coefficient for book-to-market. This collider bias is subtle: it does not cause multicollinearity and it reduces standard errors, however it distorts inference. Two aspects make colliders particularly dangerous for investors: (a) they can change the sign of estimated coefficients, thus inducing investors to buy securities that should be sold, and to sell securities that

² This is sometimes also referred to as “omitted variable bias” in the econometrics literature.

should be bought;³ and (b) the non-causal association created by a collider cannot be monetized, because by the time the collider is observed, the value of the dependent variable is already set. The effect estimated in the regression and the performance simulated in the backtest are a mirage.⁴ See section 4 for numerous examples.

The standard two-pass procedure is particularly vulnerable to collider bias. Researchers' inclusion or exclusion of controls is typically justified on statistical criteria (e.g., increasing R-squared) rather than causal logic, see Fama and French [1993, 2015]. Similarly, the three-pass factor regression approach in Giglio and Xiu [2021] applies PCA to the matrix of asset returns to extract latent variables—i.e., directions of maximal variance in returns unexplained by observed factors—, assuming without causal evidence that those latent variables must be confounders. The problem is, PCA attempts to maximize explained variance without differentiating between confounders and colliders. As a result, model specifications that look compelling in-sample will often introduce strong biases in the estimated coefficients.

Standard model evaluation metrics—such as adjusted R-squared, AIC, BIC, and t-statistics—reward misspecification and penalize parsimony, even when the extra variables introduce collider bias. In a world of limited data and noisy signals, these practices create an illusion of robustness and profitability. Also, it is important to recognize that multiple testing adjustments do not correct for model misspecification: a model can be misspecified after a single test.

This is the essence of the factor mirage: a model that is methodologically correct by current econometric standards but fails to capture the underlying causal structure. It performs well in backtests and cross-validation but delivers disappointing results out-of-sample and in live trading, because non-causal associations are not monetized.

4. EXAMPLE: COLLIDERS AMONG BARRA FACTORS

To illustrate the above points, we apply the PC causal discovery algorithm of Spirtes et al. [2000] to the time series of daily returns for the risk factors of 85 Barra risk models. This algorithm finds the network of dependencies between variables, which are visualized in the form of causal graphs (one causal graph for each model).⁵ Figure 1 aggregates the resulting causal graphs by retaining the edges present in at least a third of the graphs. Figure 2 derives the corresponding adjacency matrix. The two entries below the main diagonal indicate for what edges the PC algorithm was not able to determine the direction of the causal relationship, namely (“size”, “liquidity”) and (“size”, “resvol”).

³ These are not hypothetical concerns. Shanken [1992] discussed the consequences of estimation error in factor betas. Giglio and Xiu [2021] showed that many popular factors are likely mispriced due to omitted variables, and introduced an intermediate principal components analysis (PCA) step for identifying potential latent confounders. While these considerations partially address concerns regarding confounder bias, the problem of collider bias has received virtually no attention in the finance literature.

⁴ For a review of backtesting methods, see Joubert et al. [2024].

⁵ A causal graph is a mathematical graph that connects two variables when one is a function of the other. The arrow starts in the cause, and points to the variable that is dependent (the variable where the effect is observed). Causal graphs allow us to visualize the structure of dependencies in a system, and identify the correct controls in an experiment. For a discussion of causal discovery algorithms, see Glymour et al. [2019] and Olivetti et al. [2025].

With the help of this discovered graph, an investor can formulate the correct model specification. In particular, an investor aiming to invest in one of those factors should condition on the ancestors of that factor, and not on its descendants, to avoid the risk of controlling for a collider. For instance, to invest on “growth”, we should control for “momentum”, “leverage”, “long-term reversal”, “profit” and “investor quality”, but not on value factors (“earnings yield”, “book-to-price”, “earnings quality”, “dividend yield”). Figure 3 highlights in green the correct controls, in red the controls to be avoided, and in grey irrelevant variables.

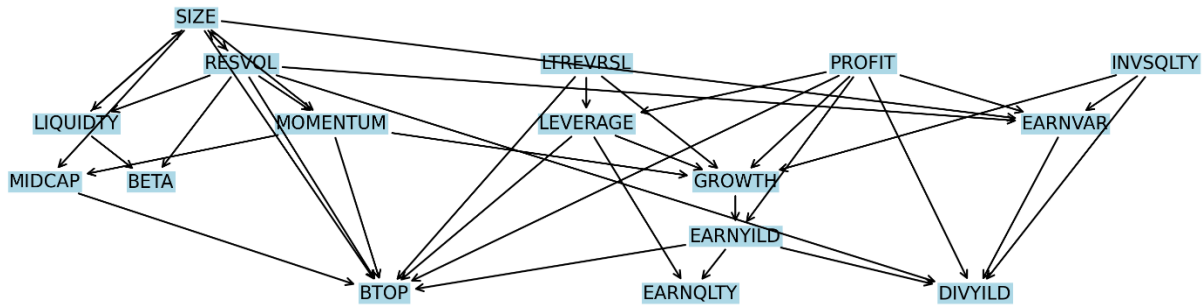


Figure 1 – Aggregate causal graph discovered through the PC algorithm

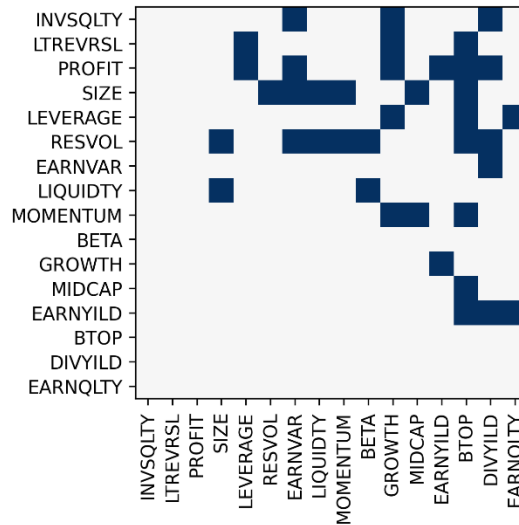


Figure 2 – Adjacency matrix

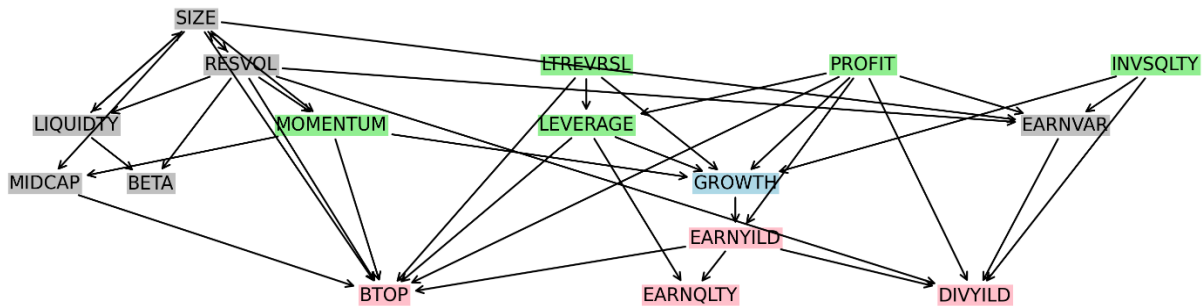


Figure 3 – Correct and incorrect controls of a Growth factor model

If we take one of those risk models, say USE4L, we can try to forecast “growth” from the other factors, selecting them to maximize the adjusted R-squared. The left plot in Figure 4 shows the adjusted R-squared of models that include all descendants, where a greedy algorithm adds ancestors –in the y-axis– one by one. The full model, with all descendants and the ancestors listed in the y-axis has an adjusted R-squared of approximately 8.5%. These models are misspecified, as descendants should not have been included as control variables. The right plot in Figure 4 shows models without descendants, where a greedy algorithm adds ancestors –in the y-axis– one by one. The full model, with all ancestors but no descendants, has an adjusted R-squared of approximately 7.8%, which is “worse” than the adjusted R-squared of the misspecified model.

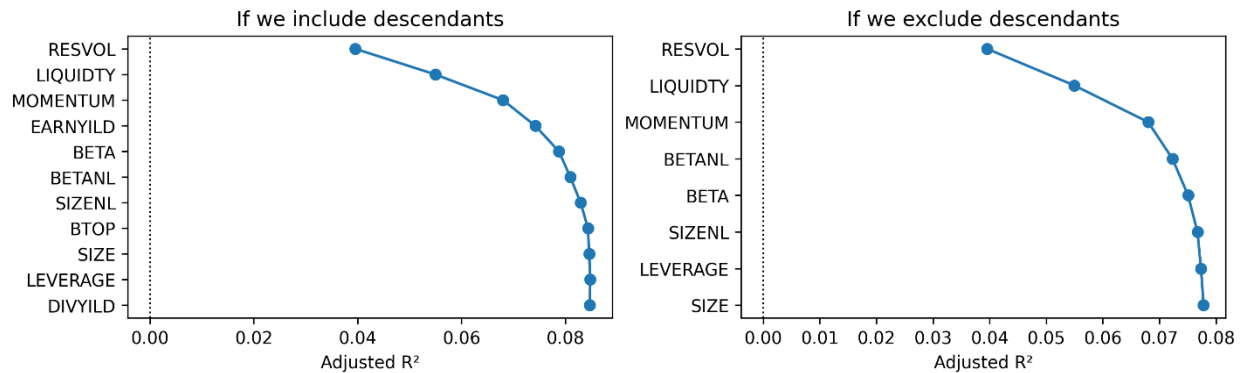


Figure 4 – Adjusted R-squared for correctly (right) and incorrectly (left) specified factor models

Model	Factor	Cause	Correct Beta	Correct aR2	Collider	Biased Beta	Biased aR2
INE2L	LIQUIDTY	RESVOL	0.0824	0.0070	BETA	-0.0362	0.0769
INE2S	LIQUIDTY	RESVOL	0.0824	0.0070	BETA	-0.0362	0.0769
THE2L	LIQUIDTY	RESVOL	0.0110	0.0000	BETA	-0.0231	0.0085
THE2S	LIQUIDTY	RESVOL	0.0110	0.0000	BETA	-0.0231	0.0085
CLE2L	EARNYILD	GROWTH	0.0817	0.0964	EARNQLTY	-0.0280	0.2316
CLE2S	EARNYILD	GROWTH	0.0817	0.0964	EARNQLTY	-0.0280	0.2316
OME2L	MIDCAP	MOMENTUM	0.0159	0.0005	BTOP	-0.0014	0.0112
OME2S	MIDCAP	MOMENTUM	0.0159	0.0005	BTOP	-0.0014	0.0112
IDE2L	EARNVAR	PROFIT	-0.0050	0.0777	DIVYILD	0.0129	0.0930
IDE2S	EARNVAR	PROFIT	-0.0050	0.0777	DIVYILD	0.0129	0.0930
MLE2L	LEVERAGE	LTREVRSL	0.0056	0.0088	EARNQLTY	-0.0014	0.0288
MLE2S	LEVERAGE	LTREVRSL	0.0056	0.0088	EARNQLTY	-0.0014	0.0288
NGE2L	LEVERAGE	LTREVRSL	0.0125	0.0151	BTOP	-0.0008	0.0199
NGE2S	LEVERAGE	LTREVRSL	0.0125	0.0151	BTOP	-0.0008	0.0199
AEE2L	GROWTH	MOMENTUM	0.0086	0.0326	BTOP	-0.0006	0.0502
AEE2S	GROWTH	MOMENTUM	0.0086	0.0326	BTOP	-0.0006	0.0502
EMM1L	GROWTH	MOMENTUM	0.0003	0.0006	BTOP	-0.0003	0.0036
EMM1S	GROWTH	MOMENTUM	0.0003	0.0006	BTOP	-0.0003	0.0036
AEE2L	MOMENTUM	RESVOL	-0.0064	-0.0002	MIDCAP	0.0017	0.0273
AEE2L	MOMENTUM	RESVOL	-0.0064	-0.0002	DIVYILD	0.0019	0.0020
AEE2S	MOMENTUM	RESVOL	-0.0064	-0.0002	DIVYILD	0.0019	0.0020
AEE2S	MOMENTUM	RESVOL	-0.0064	-0.0002	MIDCAP	0.0017	0.0273
MXE2L	GROWTH	LEVERAGE	-0.0021	0.0238	EARNQLTY	0.0032	0.0242
MXE2S	GROWTH	LEVERAGE	-0.0021	0.0238	EARNQLTY	0.0032	0.0242
USSLOWL	GROWTH	LTREVRSL	0.0022	0.0589	DIVYILD	-0.0009	0.0804
USSLOWS	GROWTH	LTREVRSL	0.0022	0.0589	DIVYILD	-0.0009	0.0804

Figure 5 – Examples of Barra risk models where adding a collider changes the sign of the estimated exposures

Figure 5 shows 26 factor models where adding a collider changes the sign of the estimated coefficient. The “Factor” column shows the explained risk factor, the “Cause” column shows a correct control, and the “Correct Beta” and “Correct aR2” columns show the estimated coefficient and adjusted R-squared when the model applies the correct control. Adding as a control the variable listed under the “Collider” column increases the adjusted R-squared as shown in the “Biased aR2” column, but it also changes the sign on the estimated coefficient as shown in the “Biased Beta” column. This list is not exhaustive and presented only for illustrative purposes.

This finding confirms that using three-pass regression approaches and black-box machine learning models is dangerous. Any choice of controls should always be argued through a causal graph, and supported by empirical evidence that those controls do not include a collider. Furthermore, the prevalence of financial colliders provides an additional argument against the “Virtue of Complexity” approach (the notion that massively overfit financial models perform better than the slightly overfit ones), see Berk [2023], Buncic [2025], Cartea et al. [2025], Fallahgoul [2025], and Nagel [2025]. In other words, given the ubiquity of colliders in financial datasets, it is generally better to underfit than overfit—if you must err, err conservatively. A slightly underfit model may still generalize reasonably, its backtested performance is less likely to be a mirage, and it will be more robust to parameter shifts.

5. BEST PRACTICES FOR PROFESSIONALS AND ASSET OWNERS

We propose the following checklist that practitioners and managers can use to assess whether a factor investing proposal is supported by causal evidence. These questions can form part of a due diligence questionnaire (DDQ), investment memo, or strategy approval process. For a detailed description of these steps, see López de Prado and Zoonekynd [2024].

Step 1: Variable Selection

- What is the intended purpose of the factor model—risk attribution or risk premia harvesting? Are the selected variables consistent with this purpose?
- How were the candidate variables initially selected?
- Were non-parametric or machine learning methods used to detect relationships?
- Were Shapley values, mean decrease impurity, or feature importance used?
- Were domain-specific constraints applied to exclude spurious or uninterpretable variables?

Step 2: Causal Discovery

- Did the researcher construct a causal graph to represent the structure of the problem?
- Were causal discovery algorithms (e.g., PC, LiNGAM, GES) used? If so, which ones?
- What economic rationale or domain expertise supports the chosen graph?
- Have alternative causal graphs been considered and ruled out? On what basis?
- Are the causal graph’s assumptions clearly documented and available for review?
- Has the causal graph structure been tested, using DoWhy’s “refutations” functionality?

Step 3: Causal Adjustment

- What method was used to identify the adjustment (e.g., backdoor, front-door, IV)?
- Which variables are being controlled for, and why?

- Did researchers confirm the validity of the adjustment using do-calculus software, such as Dagitty⁶ or Microsoft's DoWhy or networkx's `is_d_separator` function?
- Were any known colliders included in the model? How were they identified and excluded?
- Are all control variables economically interpretable and justifiable?

Step 4: Causal Explanatory and Predictive Power

- Does the model allow the simulation of controlled experiments? Does it answer counterfactual questions?
- How was the model's generalization error estimated? Is this approach realistic, given the causal graph?
- Was the model's performance assessed in terms of: (a) Probability of return sign? (b) Ranking of returns? (c) Magnitude of returns?
- Were the findings robust across multiple validation techniques or subsamples?
- Does the model show signs of overfitting to any specific performance metric?
- What is the strength and robustness of the causal effect? How strong should a missing confounder be to change the sign of the effect? Does removing a supposed confounder change the sign of the effect?

Step 5: Causal Portfolio Construction

- How are causal effects translated into portfolio weights?
- Does the strategy remain neutral or agnostic to non-causal factors?
- Is the causal graph used to hedge unwanted exposures or to guide hedging?
- Are stress tests performed on the causal model (e.g., DAG perturbation)?
- What portfolio constraints and transaction costs are considered, and how are they incorporated?
- Is the transfer coefficient between the ideal and actual portfolio computed and reported?

Step 6: Backtesting Methodology

- Which backtest methods were used (walk-forward, resampling, Monte Carlo)?
- Was the causal graph used to simulate scenarios?
- Are limitations of historical resampling or burn-in periods acknowledged?
- Are multiple scenarios or paths evaluated, not just one historical realization?
- Was combinatorially-purged cross-validation or a similar technique used to estimate a distribution of Sharpe ratios?

Step 7: Multiple Testing Adjustment

- Are all models or hypotheses tested during research available to peer reviewers?
- Was the effective number of tests estimated, accounting for test correlation?
- Were *p*-value adjustments (e.g., Holm, Hochberg, Benjamini-Hochberg) applied?
- Was the Deflated Sharpe Ratio (DSR) reported? If so, how was it computed?
- Are all statistical significance levels adjusted for the model selection process?

Final Assessment

- Is the entire research process transparent, documented, and reproducible?

⁶ See <http://www.dagitty.net/>

- Are assumptions stated explicitly and subject to falsifiability or peer-review?
- Has the model been stress-tested against changes in its structure or estimation method?
- Does the investment team demonstrate familiarity with causal inference concepts and limitations?

By insisting on these standards, asset owners and supervisors can protect capital, enhance accountability, and improve alignment between strategy design and economic intent.

6. THE ECONOMIC COST OF CAUSAL NEGLECT

Misapplying associational tools to causal problems is not just a technical misstep—it has real-world consequences. Portfolios are constructed, risks are hedged, and billions of dollars are deployed based on models that are misspecified.

The economic costs of causal neglect fall into several categories:

- Capital misallocation: Investors allocate to strategies that appear statistically significant but are not economically meaningful. This misdirection persists until performance disappoints or capital is withdrawn (López de Prado et al. [2025]).
- Hidden leverage and risk stacking: When multiple models share similar specification errors, portfolios may unknowingly stack risk exposures. For example, many value strategies may be exposed to the same macroeconomic confounders (López de Prado [2023]).
- Excessive turnover: Spurious signals lead to unnecessary trades, increasing transaction costs, bid-ask spreads, and slippage. This further erodes alpha, to the benefit of competitors.
- Lack of persistence: Models built on non-causal relationships often fail to persist when economic conditions change or new data becomes available. A shift in a collider’s parameter can spuriously flip the sign of the estimated risk premium. This explains the so-called “time-varying risk premia” conundrum. Investment models based on causal relationships can be profitable even if parameters shift (López de Prado and Zoonekynd [2024]).
- Loss of trust: When backtests consistently outperform live performance, clients lose trust in academic work and the scientific validity of systematic investing. This reputational damage impacts academics and practitioners alike, and it is difficult to reverse (López de Prado [2015]).

In short, causal neglect leads to inefficient investment decisions, unrewarded risks, and poor stewardship of capital. The current state of factor investing—marked by underwhelming performance, crowded trades, and skepticism—can be traced, in part, to these methodological shortcomings.

7. CONCLUSION

Investment factors exist, but the way researchers build and evaluate them is flawed. The econometric canon’s neglect of causality has led to a proliferation of anomalies, most of which fail to hold up under scrutiny or deliver in practice. This phenomenon, which we call the factor mirage, reflects the consequences of model misspecification—particularly collider bias and confounder bias—within canonical estimation frameworks.

Our goal is not to impose a rigid orthodoxy but to call for better causal reasoning. Causal reasoning has a long tradition in economics, however it has largely been ignored in finance. Fields like medicine, policy evaluation, and macroeconomics have already embraced causal inference with measurable benefits. It is time for asset management to evolve. The transition from associational to causal factor modeling will not be easy. It requires rewriting textbooks, questioning accepted dogmas, upgrading academic programs, unlearning old habits, revisiting familiar practices, and retooling teams. But the rewards are worth it: more stable strategies, better risk control, clearer communication, and ultimately more trustworthy products.

In a world increasingly skeptical of backtests and statistical alchemy, causal factor investing offers a credible path forward. It replaces the illusion of precision with the discipline of structure, helping us see beyond the mirage.

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